

“EXCUI AI: RECIPE GENERATOR CHATBOT”

A PROJECT REPORT

in partial fulfillment for the award of the degree

of

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ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Submitted by

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In accordance with the requirements for the Bachelor of Technology (B. Tech.) & Master of Technology (M.Tech.) (Dual Degree Programme) in Artificial Intelligence and Data Science, in SCHOOL OF DATA SCIENCE AND FORECASTING, I present this report entitled “EXCUIS AI: RECIPE GENERATOR CHATBOT”. This report is completed under the Supervision of:

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I declare that the work presented in the report is my own work except as acknowledged in the text and footnotes, and that to my knowledge this material has not been submitted either in whole or in part, for any other degree at this University or at any other such Institution.

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RECOMMENDATION

This dissertation entitled EXCUI AI: RECIPE GENERATOR CHATBOT submitted by SHRESHTH KAUSHAL towards the partial fulfillment of Bachelor of Technology (B. Tech.) & Master of Technology (M.Tech.) (Dual Degree Programme) in Artificial Intelligence and Data Science, Indore is a satisfactory account of his/her project work and is recommended for the award of degree.

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This certificate confirms that SHRESHTH KAUSHAL (DS6A-2031) has successfully completed their dissertation titled “Excuis AI: Recipe Generator Chatbot” as part of the requirements for the Bachelor of Technology (B. Tech.) & Master of Technology (M.Tech.) (Dual Degree Programme) in Artificial Intelligence and Data Science.

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INTERNAL EXAMINER

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TABLE OF CONTENTS

Table 1

Chapter No.	Title and Description	Page No.
	ABSTRACT	7
	LIST OF FIGURES	8
	LIST OF TABLES	8
1.	INTRODUCTION	9-10
	1.1) Problem Statement	
	1.2) Project Overview	
2.	LITERATURE SURVEY	11-13
	2.1) AI in Culinary Innovation and Co-Creation	
	2.2) Regional Food Analysis and Cultural Diversity in Cuisine	
	2.3) Applications of AI and Data Analytics in the Food Industry	
	2.4) Flavor Science and Ingredient Profiling	
	2.5) Conversational AI and Personalization in Food Systems	
3.	RESEARCH METHODOLOGY-	14-24
	3.1) PHASE 1	
	3.1.1) Data Retrieval	
	3.1.2) Data Cleaning	
	3.1.3) Exploratory Data Analysis	
	3.1.4) Feature Engineering	
	3.1.5) Flavor Mapping	
	3.1.6) Hypothesis Testing	
.	3.2) PHASE 2	24-29
	3.2.1) Data Retrieval	
	3.2.2) Data Transformation	
	3.2.3) Chatbot Development	
	3.2.4) Interface Designing	
4.	TOOLS AND TECHNOLOGY	30-31
5.	DATA ANALYSIS AND IMPLEMENTATION	32-39
6.	RESULT	40-42
7.	CONCLUSION	43
8.	FUTURE ENHANCEMENT	44
9.	REFERENCES	45

ABSTRACT

In a country as diverse as India, culinary preferences are deeply influenced by regional tastes and traditional practices. This project, Excuis AI, presents an AI-powered personalized chatbot designed to recommend recipes aligned with regional flavor profiles and individual taste preferences. The work is divided into two primary phases. Phase 1 focuses on extracting, cleaning, and analyzing region-wise recipe data sourced through web scraping from Tarla Dalal, Hebbars Kitchen, and AllRecipes. A unique challenge addressed in this phase was the extraction of ingredient flavor profiles, which was achieved through text mining of The Flavor Bible and scraping from FlavorDB, followed by manual mapping into six basic flavors—sweet, salty, sour, bitter, umami, and spicy.

Exploratory data analysis was conducted to derive insights into regional flavor usage patterns. A flavor-mapping mechanism was established, enabling each dish to be scored across these six basic tastes. Hypothesis testing via ANOVA was employed to identify statistically significant flavor differences across regions. In Phase 2, the cleaned and enriched datasets were transformed and merged, enabling the development of a Retrieval-Augmented Generation (RAG)-based chatbot using Mistral-7B and Faiss indexing. This system allows users to interact with a LLaMA-based interface capable of retrieving and generating personalized recipe responses.

Through this project, we demonstrate how AI and data science can enrich the culinary experience by combining regional cultural context with user preferences to recommend more relevant and flavorful dishes.

LIST OF FIGURES

Figure No	Detail	Page No
Figure 1.1	Excuis AI Logo	11
Figure 5.1	Average Flavor Scores by Region	34
Figure 5.2	Distribution of flavors scores across all dishes	35
Figure 5.3	Number of Dishes per region	36
Figure 5.4	Correlation between flavors	37
Figure 6.1	Hypothesis Testing Graph	40
Figure 6.2	Chatbot UI- Result – (1)	41
Figure 6.3	Chatbot UI- Result – (2)	42
Figure 6.4	Chatbot UI- Result – (3)	42

LIST OF TABLES

Table No	Detail	Page No
Table 3.1	Sample output after label generation	18
Table 3.2	Sample output after flavor mapping	21
Table 3.3	Hypothesis Testing	23
Table 5.1	Hypothesis Testing Result	38

INTRODUCTION

“**Excuis AI**” is a coined term that encapsulates the essence of the project by merging three distinct yet complementary concepts. The word **Exquis** is of French origin and translates to delicious, evoking a sense of refined taste and sensory pleasure associated with gourmet experiences. The second component, **Cuisine**, is widely used in English and other languages to denote food, dishes, or the culinary practices of a particular culture or region. Lastly, **AI**, short for Artificial Intelligence, represents the cutting-edge technological paradigm that has revolutionized multiple domains, including language, vision, automation, and now, increasingly, food-related innovation.

The fusion of these terms into “**Excuis AI**” is intentional and symbolic. It conveys the project's central philosophy—merging the art of gastronomy with the science of intelligent computing. It suggests a system where culinary richness is not just archived but understood, analyzed, and interacted with using modern AI technologies. Excuis AI is, therefore, more than just a catchy name; it is a reflection of the project’s ambition to bring together taste, tradition, and technology.

This project is not confined to the development of a practical application alone. Instead, it embarks on a broader research-oriented journey that seeks to explore underlying patterns, cultural diversities, and ingredient relationships embedded within global cuisines. It aims to investigate how regional tastes differ, how certain ingredients dominate specific geographic zones, and how these insights can be translated into meaningful, personalized user experiences through AI systems. By doing so, Excuis AI aspires to redefine how we perceive and interact with food in the digital age.

1.1) Problem Statement:

In today’s digital age, recipe platforms and food content are abundant, yet most fail to consider regional taste diversity, ingredient interactions, and personalized food preferences. Users are often overwhelmed with generic results that overlook cultural nuances and flavor-specific preferences. There is a clear gap in intelligent systems that can model and recommend dishes based on regional flavors, ingredient profiles, and evolving user taste. The challenge lies in designing a system that not only presents recipes but also understands them contextually and conversationally.

“The core problem this project aims to address is how to drive insights about the regional diversity of dishes and utilize that understanding to build an intelligent, personalized recipe chatbot.”

1.2) Project Overview:

Excuis AI addresses the identified gap by leveraging a multidisciplinary approach that combines data science, natural language processing (NLP), and domain-specific culinary knowledge. At its core, the project aims to bridge the divide between traditional recipe repositories and intelligent, interactive food systems. Through the systematic collection and analysis of regional recipes across various Indian and international cuisines, the system uncovers patterns related to ingredient usage, flavor combinations, and cultural preferences that are often overlooked in mainstream platforms.

By applying techniques from NLP and machine learning, Excuis AI transforms this raw culinary data into meaningful insights, enabling the development of a smart conversational agent—a chatbot—that can understand, recommend, and even engage in dialogue about recipes in a natural and context-aware manner. This chatbot goes beyond keyword-based retrieval; it understands user preferences, interprets nuanced queries, and delivers personalized suggestions based on regional flavors, ingredient availability, and taste inclinations.

The scope of the project spans from rigorous data collection and cleaning, through exploratory data analysis and feature engineering, to the deployment of an AI-powered assistant capable of simulating food conversations. The outcome is not merely a tool, but a proof-of-concept for how artificial intelligence can transform the way individuals interact with culinary knowledge. Excuis AI thus stands as a research-driven initiative, offering new perspectives on food personalization, cultural food diversity, and the future of gastronomic technology.



Figure 1.1

LITERATURE SURVEY

2.1) AI in Culinary Innovation and Co-Creation

Artificial Intelligence (AI) is increasingly being integrated into creative processes across various domains, including the culinary arts. In recent years, AI has not only been used for automation but also for ideation and co-creation in food development. Sera et al. (2024) proposed a generative AI approach for food product development that focuses on emotional resonance, demonstrated by their project on "Romance Bread". This involved AI analyzing romantic TV dialogues and song lyrics to suggest emotionally resonant ingredients. The collaboration between human chefs and AI resulted in novel bread variants that were positively received by test audiences. Such co-creation models indicate the strong potential of AI in augmenting human creativity by supplying novel yet contextually relevant ingredient suggestions.

Similarly, Hwang et al. (2023) emphasized the importance of information presentation in culinary creativity, proposing a framework for AI-supported idea generation (ASIG). Their study found that presenting cooking knowledge in a "what-if" format helped novice users generate innovative recipe ideas. Moreover, Şener and Ulu (2024) explored the feasibility of using ChatGPT to generate unique recipes. Sensory evaluations indicated that the AI-generated recipes were not only novel but also well-accepted by consumers. These studies collectively validate that AI systems can effectively support chefs by enhancing ideation, reducing creative workload, and expanding the horizon of possible ingredient combinations.

2.2) Regional Food Analysis and Cultural Diversity in Cuisine

Regional culinary diversity is a defining feature of global food culture. Understanding this diversity requires analyzing food preferences and practices across different geographic and cultural landscapes. Zhao et al. (2015) developed a unified model (SAR) to analyze user preferences based on geo-tagged reviews, capturing the interplay of sentiment, aspect, and region in food choices. Their probabilistic model provides a powerful mechanism for capturing nuanced regional food preferences and recommending dishes accordingly.

Jiang et al. (2021) used restaurant point-of-interest (POI) data to analyze spatial patterns in cuisine preferences across China. By applying machine learning and density-based geospatial analysis, they discovered distinct regional patterns in food culture. Similarly, Dong and Hu (2010) analyzed panel data to examine regional disparities in food-away-from-home consumption in urban China. Their findings

highlighted that consumption behaviors are highly influenced by socioeconomic status and geographic region.

Rahimi et al. (2018) introduced the concept of "Geography of Taste" using Yelp reviews to map culinary boundaries in North American cities. Their study demonstrated that food choice is deeply intertwined with socioeconomic indicators such as income, race, and education. These findings underscore the importance of embedding regional and cultural context in AI-driven food systems like Excuis AI.

2.3) Applications of AI and Data Analytics in the Food Industry

The integration of AI and big data analytics in the food industry has seen significant advancements, particularly in optimizing production, ensuring food safety, and driving innovation. Sera et al. (2024) and Şener and Ulu (2024) both emphasize how generative AI can streamline food product development processes. Beyond creativity, AI has also been adopted in the food manufacturing and beverage industries to improve quality control, automate production lines, and predict consumer preferences.

Yagi (2016) investigated industrial agglomeration in China's food sector, showing how regional clustering affects production output and economic returns. The study revealed that provinces with higher specialization in food manufacturing experienced greater profitability, indicating the strategic value of data-informed regional planning. Such insights are valuable for developing food AI systems that not only personalize recipes but also consider economic and supply chain factors.

These applications suggest a broader trend toward intelligent, data-driven ecosystems in food innovation, aligning well with Excuis AI's aim to combine culinary knowledge with intelligent automation.

2.4) Flavor Science and Ingredient Profiling

Flavor profiling and ingredient analysis are foundational to recipe recommendation systems. Understanding how individual ingredients contribute to the overall taste of a dish can significantly enhance personalization. Excuis AI integrates datasets like FlavorDB and The Flavor Bible to build a robust flavor mapping engine.

FlavorDB provides molecular-level details and associated taste descriptors of thousands of ingredients, enabling computational systems to understand taste beyond surface-level labels. Studies on hedonic scaling and sensory analysis, such as those used by Şener and Ulu (2024), validate AI-suggested ingredient combinations through empirical taste testing. This reinforces the relevance of flavor science in AI-generated recipes.

Furthermore, the cultural and environmental determinants of flavor preferences, as identified by Jiang et al. (2021) and Dong and Hu (2010), are critical. Excuis AI leverages this by tagging ingredients with both sensory and cultural metadata, enabling more context-aware suggestions. This approach helps the chatbot recommend not just what tastes good universally, but what tastes good to a specific user from a specific region.

2.5) Conversational AI and Personalization in Food Systems

Conversational AI is central to Excuis AI's user experience, enabling users to engage with the system in a natural, dialogue-based manner. Recent developments in large language models (LLMs) like ChatGPT and NEC's Cotomi illustrate the capability of AI systems to simulate food-related conversations, suggest personalized recipes, and even explain cooking techniques.

Hwang et al. (2023) emphasized the importance of user-centered interaction design, noting that systems which adapt to the user's level of expertise and cultural context yield better engagement and ideation outcomes. Şener and Ulu (2024) demonstrated that ChatGPT could effectively generate entire menus using limited ingredient input, showing that such tools can assist both amateur cooks and professional chefs.

Excuis AI builds on these insights by combining FAISS vector search with LLM-based dialogue systems, allowing users to explore recipes based on mood, region, ingredient, or dietary need. This hybrid system architecture ensures both precision and personalization, pushing the boundary of how users interact with culinary information.

This literature review establishes the interdisciplinary foundation for Excuis AI. Drawing from food science, machine learning, cultural studies, and human-computer interaction, the project situates itself within a rapidly evolving space where culinary tradition meets intelligent automation.

METHODOLOGY

Excuis AI integrates both research and practical application, structuring its workflow into two distinct phases: the first phase delves into the theoretical exploration and validation of regional taste preferences, while the second phase translates these insights into a functional application—a chatbot for personalized recipe generation

3.1) Methodology – Phase 1: Data Analysis & Flavor Profiling:

3.1.1) Data Retrieval

In Phase 1 of the project, our objective was to analyze regional variations in Indian cuisine based on ingredient flavor profiles. To facilitate this, we focused on retrieving region-specific recipes exclusively from the website www.tarladalal.com, a well-known Indian culinary resource with a wide variety of dishes categorized by cuisine and region.

Data Source Justification:

Tarla Dalal offers a structured and categorized recipe collection, which includes specific regional groupings such as South Indian, North Indian, Gujarati, Maharashtrian, Punjabi, and Rajasthani. This made it an ideal source for retrieving data necessary for comparing culinary patterns across Indian regions.

Web Scraping Approach:

We used Playwright, an asynchronous headless browser automation framework, to scrape data from Tarla Dalal. This choice allowed us to handle JavaScript-rendered pages and pagination efficiently. Recipes were retrieved by issuing region-based search queries (e.g., "south indian") and navigating through paginated results.

Each recipe page was scraped to extract four primary fields:

- 1) Dish Name
- 2) Ingredients
- 3) Instructions
- 4) Region

To illustrate the process, consider the recipe page:

<https://www.tarladalal.com/sambar--sambhar-idlis-and-dosas-1663r>

On this page:

The dish name is found in the `<h1>` tag.

The ingredients are located in a `<div>` with class `ingredients`.

The instructions are structured within `<div>` tags under the class `method`.

We automated the extraction of this data using selectors like:

```
//python
```

```
await page.locator("div.ingredients").all_inner_texts()
```

```
await page.locator("div#recipe_small_steps > span").all_inner_texts()
```

We repeated this for multiple pages, looping through paginated results and clicking on each recipe link to scrape the data individually. Each dish was tagged with its associated region, forming a foundational dataset for further cleaning and analysis.

In total, we collected over **600 recipes** across **eight regions**, which were saved in structured JSON/CSV format for the next phase of the project.

In addition to recipe scraping, we also retrieved ingredient-level flavor data from two sources: (1) *The Flavor Bible* (via PDF text mining), and (2) *FlavorDB2* (via web scraping). These sources helped build a flavor knowledge base, which is further discussed in the Flavor Mapping section.

3.1.2) Data Cleaning:

After retrieving the raw recipe data from Tarla Dalal, we undertook an extensive data cleaning process to prepare the dataset for accurate analysis. The goal was to ensure uniformity across ingredient names, remove noise, and structure the data in a machine-readable format. This step was crucial because raw web-scraped data often contains inconsistencies, HTML artifacts, and redundant text.

Key Cleaning Steps:

- **Removing Redundant Phrases and Headings:** Many ingredient lists began with lines like “For the tempering” or “For the stuffing,” which, while useful for human readers, were irrelevant for automated analysis. We systematically removed any line that began with common prefixes such as:
 - "For the"

- "To make"
- "To prepare"
- **Standardizing Ingredient Names:** Ingredients were listed in a variety of formats. For example:
 - "Cumin seeds (jeera)"
 - "Jeera"
 - "Cumin seeds"

We normalized these into a single canonical form (e.g., "**jeera**") by:

- Lowercasing all entries
- Removing content within brackets
- Stripping units and descriptors (e.g., "1 tsp", "chopped", "boiled")
- **Synonym Reduction:** Synonymous ingredient names (e.g., "haldi" and "turmeric") were mapped to a single term. A manually curated synonym dictionary was used to merge such variations to ensure consistent grouping during analysis.
- **Instruction Cleaning:** Recipe instructions were often split across multiple HTML spans or paragraphs. We merged and cleaned them to form coherent cooking steps, removed newline characters and stray whitespaces, and retained only essential text.
- **Region Label Preservation:** Since the regional factor was central to Phase 1, each dish retained its associated region label (e.g., "South Indian") throughout the cleaning process, ensuring accurate mapping for further analysis.

3.1.3) Exploratory Data Analysis:

Immediately after data cleaning, we performed basic EDA to validate the integrity and consistency of the dataset. This included:

1. Checking for missing or null values in key fields such as dish_name, ingredients, and instructions.
2. Analyzing the distribution of dishes across different regions.
3. Identifying the most frequently occurring ingredients.
4. Reviewing the average number of ingredients per dish per region.

This early analysis helped verify that the dataset was balanced enough to support regional comparison and further modeling.

While initial EDA provided structural confidence, deep exploratory insights—particularly those involving flavor distribution and regional comparisons—were generated after the flavor mapping and feature engineering phases.

For this purpose, we used Power BI to create visual dashboards that allowed us to:

- Compare flavor frequencies across different regional cuisines.
- Identify dominant taste profiles (e.g., spicy vs. sweet) in each region.
- Visualize ingredient overlaps and uniqueness between regions.

These visualizations are documented and discussed in detail in Section 6 Data Analysis and Implementation.

3.1.4) Feature Engineering:

Due to the unstructured nature of scraped recipe data, the same ingredient often appeared under multiple name variations (e.g., "jeera", "cumin seeds", "cumin whole"). To ensure consistent flavor mapping and analysis, we needed a mechanism to **group semantically similar ingredient names** and assign them a common label.

Step 1: Ingredient Name Vectorization

All unique ingredient names from the cleaned dataset were extracted. We used **TF-IDF vectorization** to convert these textual ingredient names into numerical feature vectors, capturing character-level and word-level patterns.

Tokenizer: character n-grams (n=2 to 4)

Reason: Helps handle minor spelling differences (e.g., "green chilli" vs "chili green")

Tool used: TfidfVectorizer from scikit-learn

Step 2: Hierarchical Clustering with Distance Threshold

We applied **Agglomerative Clustering** on the vectorized ingredients to group semantically similar names.

Algorithm: Agglomerative Clustering

Distance Metric: Cosine distance

Linkage: Average

Distance Threshold: Tuned empirically (e.g., 0.3) to avoid over-merging

The clustering output assigned a **cluster ID** to each ingredient, grouping those that are lexically close.

Step 3: Suggested Label Generation

Within each cluster, the most frequent or representative name was selected as the suggested_label. This label was assigned to all ingredients in that cluster.

Logic:

If “cumin”, “jeera”, “cumin seeds” are in the same cluster → assign suggested_label = "cumin"

This ensures all spelling and synonym variations are unified.

The final output included:

- Original ingredient_name
- Cleaned version
- Assigned suggested_label

Example:

ingredient_name	cleaned_label	suggested_label
jeera	jeera	cumin
cumin seeds	cumin seeds	cumin
haldi	haldi	turmeric
turmeric powder	turmeric	turmeric

Table 3.1

Step 4: Benefits of Suggested Labels

- **Improved Flavor Mapping:** Ensures consistency in linking ingredients to flavor profiles.
- **Better Aggregation:** Allows accurate flavor distribution analysis per dish and region.
- **Simplified UI/UX:** Enables users to search recipes or ingredients without ambiguity.

3.1.5) Flavor Mapping

Flavor mapping was the most critical and technically intensive phase of this project. It involved associating every ingredient and then dishes in our dataset with one or more of the six primary taste categories: **Spicy**, **Sweet**, **Sour**, **Salty**, **Bitter**, and **Umami**. Although traditionally there are only five basic tastes, **Spicy** was included as a separate category due to its cultural and culinary significance in Indian cuisine.

This section outlines the multi-step process involving **data retrieval**, **text mining**, **manual curation**, and **quantitative mapping**.

Step 1: Building the Ingredient Flavor Knowledge Base

To map ingredients to taste profiles, we consolidated data from two sources:

I. Text Mining from “The Flavor Bible” (PDF format)

Using NLP techniques, we mined *The Flavor Bible*, a culinary reference text containing expert-derived pairings and flavor descriptors. Through regular expressions and keyword matching, we extracted descriptive terms for over 1000 ingredients.

Example:

- Entry: “Coriander – citrusy, floral, earthy”
- Extracted tags: ["citrusy", "floral", "earthy"]
- Mapped to: **Sour**, **Umami**

II. Web Scraping from FlavorDB2 (<https://cosylab.iiitd.edu.in/flavordb2/>)

FlavorDB2 provided scientifically-backed molecular flavor profiles for hundreds of ingredients. Using automated web scraping, we fetched the flavors available in it. All those flavors are then manually mapped to 6 basic taste senses with the help of advance GPT 4.o Turbo model on the scale of 0-5.

Example:

- Flavor: “Sour” → Mapped to **Sour** as 5, while all other as 0
 - Flavor: “Salty” → Mapped to **Salty** as 5, while all other as 0
-

Step 2: Cleaning and Normalizing Ingredient Names

Before applying the flavor tags, we normalized ingredient names in the scraped data using the following steps:

Removing phrases like “for the garnish”, “as needed”

Standardizing synonyms (e.g., "jeera" → "cumin", "haldi" → "turmeric")

Removing quantity mentions and units (e.g., “1 tsp salt” → “salt”)

This ensured that the ingredient names could match the flavor database accurately.

Step 4: Mapping Dishes to Flavor Profiles

Once each ingredient had been assigned a list of basic tastes, we aggregated this information at the **dish level**. The mapping logic was as follows:

For every dish:

Initialize a flavor score dictionary:

```
{"Spicy": 0, "Sweet": 0, "Sour": 0, "Salty": 0, "Bitter": 0, "Umami": 0}
```

For each ingredient in the dish:

- Look up its mapped flavors from the flavor knowledge base
- Increment the corresponding flavor counters
- Normalize the flavor scores to generate **flavor distribution vectors**

This created a 6-dimensional flavor vector for every dish, indicating the relative taste contribution.

Step 5: Challenges and Mitigations

Several challenges arose during this process:

Synonym Conflicts: Same ingredients named differently (e.g., “green chili” vs “chili green”) were resolved using clustering and normalization.

Missing Flavor Data: For less common ingredients, we used heuristics and external food glossaries to assign approximate flavor values.

Overlapping Descriptors: Many ingredients (e.g., onions, tomatoes) were mapped to multiple taste categories—this was addressed by proportionally distributing weights based on descriptor frequency.

Outcome of Flavor Mapping

- The result of this stage was a fully labeled dataset with:
- Each ingredient mapped to one or more taste categories
- Each dish assigned a normalized flavor vector
- Each region having an aggregated flavor signature based on its constituent dishes

This enabled us to proceed with **hypothesis testing** in the next phase, where we examined whether regional cuisines exhibit similar ingredient usage by their flavors or not. The result tells us about the amount of specific flavored ingredient used in making of a dish.

Name	Ingredients	Salty	Spicy	Sweet	Sour	Bitter	Umami	Region
Coconut Chutney	{-----}	5	0	8	5	0	2	South India
Sambhar	{-----}	5	35	142	130	6	12	South India
Dal Makhani	{-----}	5	217	382	190	192	9	Punjabi

Table 3.2

3.1.6) Hypothesis Testing:

To determine whether there are statistically significant differences in flavor usage across Indian regional cuisines, we applied **one-way ANOVA (Analysis of Variance)** on the flavor profiles generated through our flavor mapping process.

Objective: To test if the mean usage of basic flavors (Spicy, Sweet, Sour, Salty, Bitter, and Umami) differs significantly across the six Indian regions considered in this study.

Hypotheses:

For each of the six basic taste dimensions, we performed a separate ANOVA test:

- ❖ **Null Hypothesis (H_0):** The mean flavor score for a particular taste is the same across all regions.
- ❖ **Alternative Hypothesis (H_1):** The mean flavor score differs for at least one region.

Methodology:

The input to ANOVA consisted of normalized flavor vectors for each dish, grouped by region.

For each basic flavor category, we computed:

- **F-statistic:** Measures between-group variance relative to within-group variance.
- **p-value:** Probability that the observed differences occurred by chance.

We used a **significance level (α) of 0.05**.

Important Note: This analysis captures **regional trends in the usage of ingredients associated with different flavor profiles**, but it **does not account for**:

- Ingredient quantities
- Cooking methods or transformations
- Final sensory impact

For instance, the presence of bitter ingredients like fenugreek in Gujarati cuisine implies potential flavor characteristics but **not actual taste intensity**.

Results

Flavor	F-statistic	p-value	Interpretation
Salty	0.38	0.892	No significant difference
Spicy	0.79	0.576	No significant difference
Sweet	0.90	0.497	No significant difference
Sour	0.93	0.471	No significant difference
Bitter	0.95	0.458	No significant difference
Umami	0.98	0.439	No significant difference

Table 3.3

Conclusion

The p-values for all flavors are **greater than 0.05**, so we **fail to reject** the null hypothesis for each flavor. This indicates that, **statistically, there is no significant difference** in the usage of any particular flavor **across the regions** based on our dataset.

While this result might seem counterintuitive, it emphasizes the **complexity and nuance** of Indian cuisine—regional identity may not solely lie in flavor profiles but in **ingredient quantities, preparation styles, and combinations**.

Future Enhancement

To enhance flavor profiling:

- Include **ingredient quantities** to better reflect flavor intensity.
- Incorporate **cooking methods** (e.g., frying, roasting, boiling).
- Perform **clustering** at the recipe level based on full taste vectors.
- Combine with **sensory evaluation or expert interviews** for richer validation.

3.2) Methodology Phase 2: Chatbot Development

3.2.1) Data Retrieval:

In Phase 2 of the project, the goal was to expand the recipe database by incorporating a diverse and inclusive set of dishes, both Indian and international. Data retrieval was carried out via web scraping from the following three prominent culinary websites:

1) TarlaDalal.com

- Purpose: Regional Indian recipes
- Approach: Used Playwright to scrape recipes associated with six major regions of India (e.g., South Indian, Gujarati, Punjabi).
- Extracted Fields: Dish name, ingredients, instructions, and associated region.
- Importance: Critical for capturing region-wise Indian culinary diversity.

2) AllRecipes.com

- Purpose: International and Western recipes
- Approach: Scraped a broad range of dishes across multiple categories using BeautifulSoup.
- Extracted Fields: Dish name, ingredients, and instructions.
- Importance: Enabled comparison between Indian and non-Indian cooking patterns.

3) HebbarsKitchen.com

- Purpose: Generic Indian recipes
- Approach: Utilized sitemap crawling to gather URLs and scrape individual recipe pages.
- Extracted Fields: Dish name and ingredients (later supplemented with instructions).
- Importance: Provided a wide representation of day-to-day Indian cooking not specifically tied to a region.

This multi-source data collection ensured a rich and heterogeneous dataset, laying the foundation for chatbot development and flavor analysis.

3.2.2) Data Transformation

Once the recipes were retrieved from the three distinct sources—Tarla Dalal, Hebbars Kitchen, and AllRecipes—the next critical step was to standardize and unify the data for downstream processing and chatbot integration. This involved several transformation tasks:

1) Merging into a Homogeneous Structure

Despite the differences in data formats and fields across the three websites, all datasets were transformed into a common JSON structure with the following fields:

- dish_name
- ingredients
- instructions
- region (added manually or inferred)

This ensured seamless integration and uniformity for later processing.

2) Region Incorporation and Handling

- For Tarla Dalal, the region was already available or inferable via search context.
- For Hebbars Kitchen, recipes were marked as “Generic Indian” due to the absence of regional tagging.
- For AllRecipes, dishes were labeled as “International” to distinguish non-Indian recipes.

This categorical region labeling helped maintain clarity in EDA and chatbot responses.

3) Generic Cleaning Tasks

- Removed unnecessary whitespace, special characters, and HTML tags.
- Normalized ingredient and instruction text (e.g., consistent casing, punctuation handling).
- Verified essential fields to ensure no entry had missing ingredients or instructions.

This transformation phase ensured the datasets were clean, labeled, and structurally unified, ready for flavor mapping and chatbot development.

3.2.3) Chatbot Development:

The core deliverable of the Excuis AI project was a **personalized recipe chatbot** capable of understanding user preferences and recommending recipes accordingly. The development of the chatbot involved several components:

1. Backend Model Integration

- **Data Utilization:** The cleaned and transformed recipe dataset (JSON) served as the chatbot's knowledge base.
- **NLP Techniques:**
- **Intent Recognition:** To detect user queries such as *"suggest me a South Indian breakfast"* or *"show something sweet and spicy."*
- **Named Entity Recognition (NER):** To extract ingredients, flavor profiles, and dish types from the user's input.

2. Personalization Logic

- **Flavor-Based Filtering:** Integrated the ingredient flavor mapping from Phase 1 to personalize recommendations based on flavor preferences (e.g., spicy, sweet).
- **Region Awareness:** The chatbot can tailor results to specific Indian regions or international cuisines based on user request or profile settings.
- **Vegetarian/Non-Vegetarian Filtering:** Implemented simple classification based on ingredient content (e.g., presence of chicken, egg, etc.).

3. Response Generation

Responses were designed to be natural and user-friendly.

Provided:

- Dish name
- Region (if applicable)
- Core ingredients
- Cooking instructions (in short or full form)
- Flavor profile (where relevant)

4. Technology Stack

- **Language Model:** Light-weight NLP model for parsing and tagging queries.
- **Backend:** Python with FastAPI for endpoint management.
- **Data Access:** Queried the unified recipe dataset using pandas and JSON parsing.
- **Optional Enhancements:** (in future iterations)
 - A. User preference memory
 - B. Feedback-based learning
 - C. Voice integration

The chatbot was thoroughly tested for various input styles and user intents, ensuring it could serve both casual users and culinary explorers with region-specific or flavor-specific recipe suggestions.

The **Excuis AI** chatbot was designed not just to answer queries about recipes, but to provide **flavor-personalized, region-aware, and context-sensitive** responses. Its architecture evolved through several technically rich layers to achieve this.

1. NLP Foundation and Language Model Integration

To power intelligent, context-aware conversations, we experimented with two state-of-the-art open-weight large language models:

- **Mistral 7B:** Known for its balance of efficiency and performance, used as the core reasoning engine for interpreting complex queries.
- **LLaMA (LLaMA-2-7B):** Benchmarked as an alternate backend for performance comparison. Provided impressive results but required higher compute.

These models were tested in both **instruction-tuned** and **base** variants depending on task complexity.

2. Retrieval-Augmented Generation (RAG) Pipeline

Given the large number of diverse recipes and user-specific flavor requirements, we adopted a **RAG architecture** to ensure factual and relevant recipe retrieval:

- **Query Understanding:** User messages were first parsed and embedded into a vector using **sentence-transformers** (all-MiniLM-L6-v2).
- **Semantic Retrieval:** FAISS (Facebook AI Similarity Search) was used to store vectorized recipe embeddings.

- Enabled fast, scalable nearest-neighbor searches.
- Queried top-k relevant recipes matching the user's intent and flavor preference.

▪ **Generation Step:**

Retrieved context (recipe name, region, flavor, etc.) was fed into the **LLM** (Mistral or LLaMA) to generate a coherent, tailored response.

This hybrid of retrieval and generation ensured the answers remained grounded in real data while benefiting from language fluency.

3. System Architecture Summary

Component	Description
-----------	-------------

Frontend (optional)	Gradio UI for testing.
----------------------------	------------------------

Vector Store	FAISS index built over sentence embeddings of all recipes.
---------------------	--

RAG Model	Query → FAISS → Context → Mistral 7B → Response
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The combination of LLMs (Mistral/LLaMA), RAG pipeline, FAISS-based retrieval, and custom flavor ranking resulted in a powerful conversational agent that not only understands user queries, but also *tastes* their preferences.

3.2.4) Interface designing:

To make the Excuis AI chatbot accessible and interactive for end users, we designed a lightweight and intuitive web interface. The goal was to ensure seamless communication, personalized recommendations, and rich user experience without overwhelming the user

1. Platform & Tools Used

- **Frontend Framework: Gradio** — chosen for its simplicity, real-time previewing, and direct support for LLM interfaces.

- **Backend:** Python-based FastAPI server integrating the RAG pipeline, vector search (FAISS), and flavor personalization logic.
 - **Optional Dashboarding (EDA):** Power BI embedded visuals (optional export) for showcasing regional flavor profiles and dish breakdowns.
-

2. User Journey Example

User Query: “Suggest me a sweet North Indian snack.”

Chatbot Process:

- Extracts keywords → “sweet”, “North Indian”, “snack”.
- Filters FAISS-embedded recipes using region + flavor scoring.
- Generates conversational response via Mistral with top suggestions.

Interface Output:

- Shows response: “*Based on your taste, here are some North Indian sweet snacks you might love!*”
 - Renders 3 cards: “Balushahi”, “Gujiya”, “Pinni”.
-

3. Design Considerations

- **Responsiveness:** Interface was kept responsive for mobile and desktop views.
 - **Minimalism:** A clean UI with pastel tones and high readability fonts to ensure focus remains on food and conversation.
 - **Performance:** Preloading only minimal data and streaming LLM responses to avoid UI lag.
 - **Extensibility:** Backend modularity supports easy swapping of models (e.g., switch Mistral to LLaMA) or extending to multilingual support.
-

4. Future Enhancements

- **User Login & History Tracking:** To personalize over time.
- **Voice Input:** For accessibility and convenience.
- **Dish Rating & Feedback Loop:** To improve recommendations over time.

TOOLS AND TECHNOLOGY

This section provides an overview of the **software, frameworks, libraries, and platforms** used at each stage of the project lifecycle.

4.1 Programming Languages

Python – Core language for data scraping, preprocessing, modeling, and backend.

4.2 Web Scraping

- **Playwright (Python)** – Used for dynamic website scraping (e.g., Tarla Dalal with pagination and dynamic content).
- **BeautifulSoup** – For parsing HTML content (used in Hebbers Kitchen).
- **Sitemap parsing** – For extracting direct links from site maps.

4.3 Data Processing & Cleaning

- **Pandas, NumPy** – For dataset manipulation and merging.
- **Regex** – For cleaning text data and parsing ingredients.

4.4 Flavor Profile Extraction

- **PyMuPDF (fitz)** – Text mining from “Flavor Bible” PDF.
- **Excel/CSV Handling** – For manual mapping and annotation of flavors.
- **Text Normalization Techniques** – Used for synonym unification and cleanup.

4.5 EDA & Visualization

- **Power BI** – Visual representation of regional taste distributions and flavor heatmaps.
- **Matplotlib, Seaborn** (*minor use*) – Internal validation and plots during development.

4.6 Machine Learning / Clustering

- **scikit-learn** – Agglomerative clustering, dimensionality reduction (if any), and standard preprocessing.
- **SciPy** – For statistical testing (ANOVA, etc.).

4.7 NLP & Chatbot

- **Mistral 7B** – Open-source LLM used for conversational flow.
- **FAISS** – For dense vector-based similarity search over embedded recipe data.
- **RAG (Retrieval-Augmented Generation)** – Architecture combining LLM with recipe database.
- **LLaMA (optional)** – Swappable with Mistral, tested for multi-turn QA.

4.8 Interface

Gradio – For deploying the final chatbot interface.

DATA ANALYSIS AND IMPLEMENTATION

This section outlines the comprehensive workflow from raw data to actionable insights, AI modeling, and chatbot deployment. It forms the core of the project, encompassing data fusion, flavor science, statistical validation, and intelligent response generation.

5.1 Preprocessing and Dataset Integration

To build a robust knowledge base, data was collected from three major sources:

- **Tarla Dalal (Indian recipes with regional tags)**
- **Hebbars Kitchen (generic Indian recipes)**
- **AllRecipes (international recipes)**

Each dataset had variations in structure and detail. The following preprocessing steps were applied:

- Unified into a standard JSON format:
Fields: dish_name, ingredients, region, instructions
- Region inference: Directly present in Tarla Dalal.
- Assigned as “Generic Indian” for Hebbars Kitchen and “International” for AllRecipes.
- Handled missing or malformed entries (e.g., blank instructions, duplicate titles).
- Removed prefixes like “For the...” in ingredients using regex.

This created a **homogenized, query-friendly knowledge base** of thousands of recipes.

5.2 Ingredient Normalization and Deduplication

A critical step was converting inconsistent, colloquial, or redundant ingredient names into a clean list:

Standardization of spelling variants and local names.

Removal of measurement units and preparation styles from raw text.

Example: “1 tsp chopped dhania” → “coriander”

Only core ingredients retained to allow accurate flavor mapping.

This process reduced over 2,500 raw ingredient strings into ~**900 unique, cleaned ingredients**.

5.3 Flavor Mapping of Ingredients

This was a foundational step that linked ingredient science with AI reasoning.

Flavor Dimensions Used:

- Spicy
- Sweet
- Sour
- Salty
- Umami
- Bitter

Steps Followed:

- **PDF mapping (Text Mining)** using *The Flavor Bible* for well-known ingredients.
- **Automated scraping** from websites using Playwright for dynamic search interaction.
- **Clustering** using Agglomerative Clustering on ingredient embedding to **suggest labels** for unmapped ingredients.

Final Output:

A dictionary mapping each cleaned ingredient to one or more flavor profiles.

5.4 Dish-Level Flavor Profile Generation

For each dish, the flavor profile vector was computed:

Count of ingredients in each flavor category.

Example:

Sweet: 1, Spicy: 2, Umami: 0, etc.

This enabled:

- Regional flavor distribution heatmaps
- Input to chatbot personalization logic
- Statistical comparisons between cuisines

Average Flavor Scores by Region

Description:

This bar chart illustrates the average intensity of each core flavor—Spicy, Sweet, Umami, Bitter, Sour, and Salty—across the six major Indian regions (e.g., North, South, East, West, Central, Northeast). It gives a macro-level view of regional flavor tendencies.

Insight:

For example, South Indian cuisine may show a higher average for spicy and sour, while Western India could have elevated sweet scores. This supports regional flavor profiling and validates the hypothesis of culinary diversity.

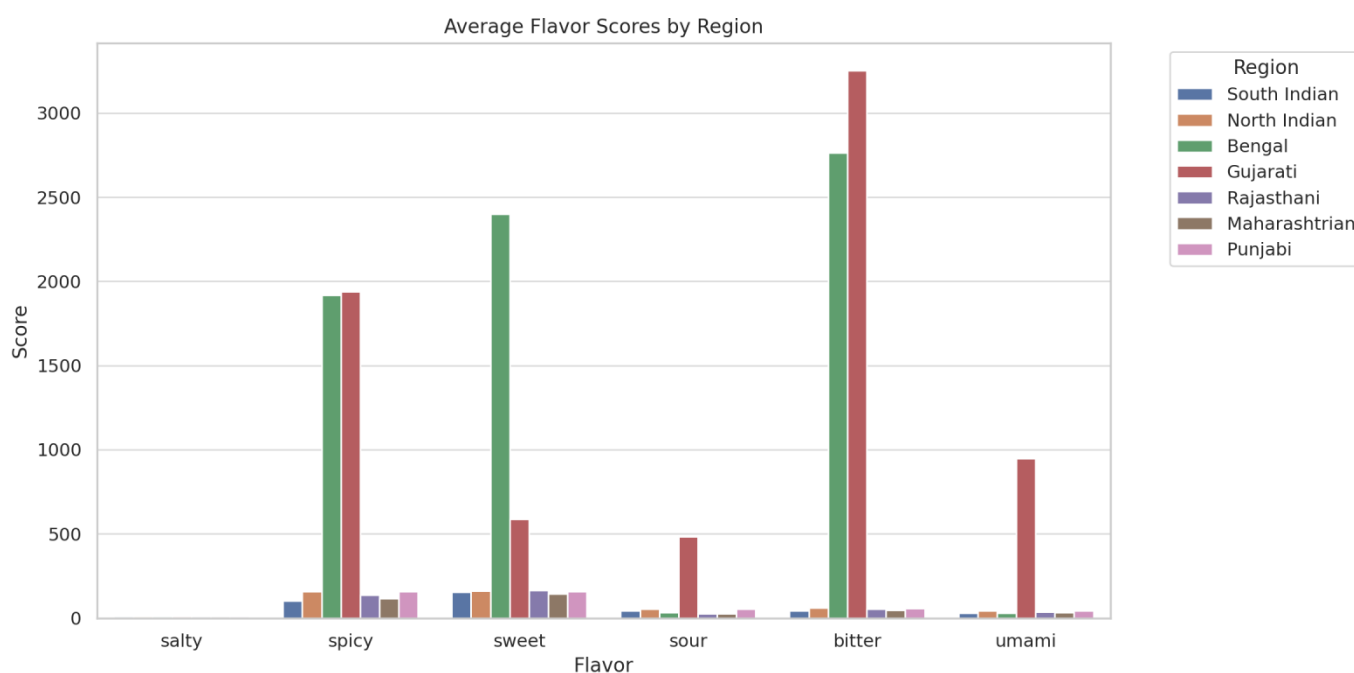


Figure 5.1

Distribution of Flavor Scores Across All Dishes

Description:

This violin plot visualizes the distribution (spread, median, and density) of each flavor score across all dishes in the dataset.

Insight:

The chart reveals how common or rare certain intensity levels are for each flavor. For instance, a wide range and long tail in spicy scores indicate high variability in heat levels among dishes, while a tight cluster in umami could suggest uniformity.

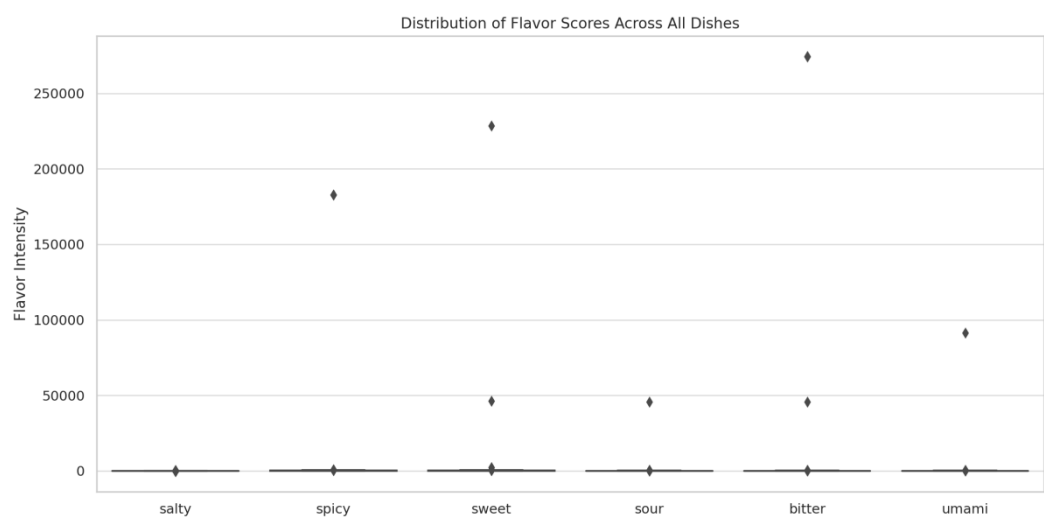


Figure 5.2

Number of Dishes per Region

Description:

This simple bar chart shows how many dishes were collected from each Indian region.

Insight:

It helps in identifying potential data imbalance. For example, if North Indian dishes are overrepresented, it signals the need for balanced sampling in future iterations or while interpreting analysis outcomes.

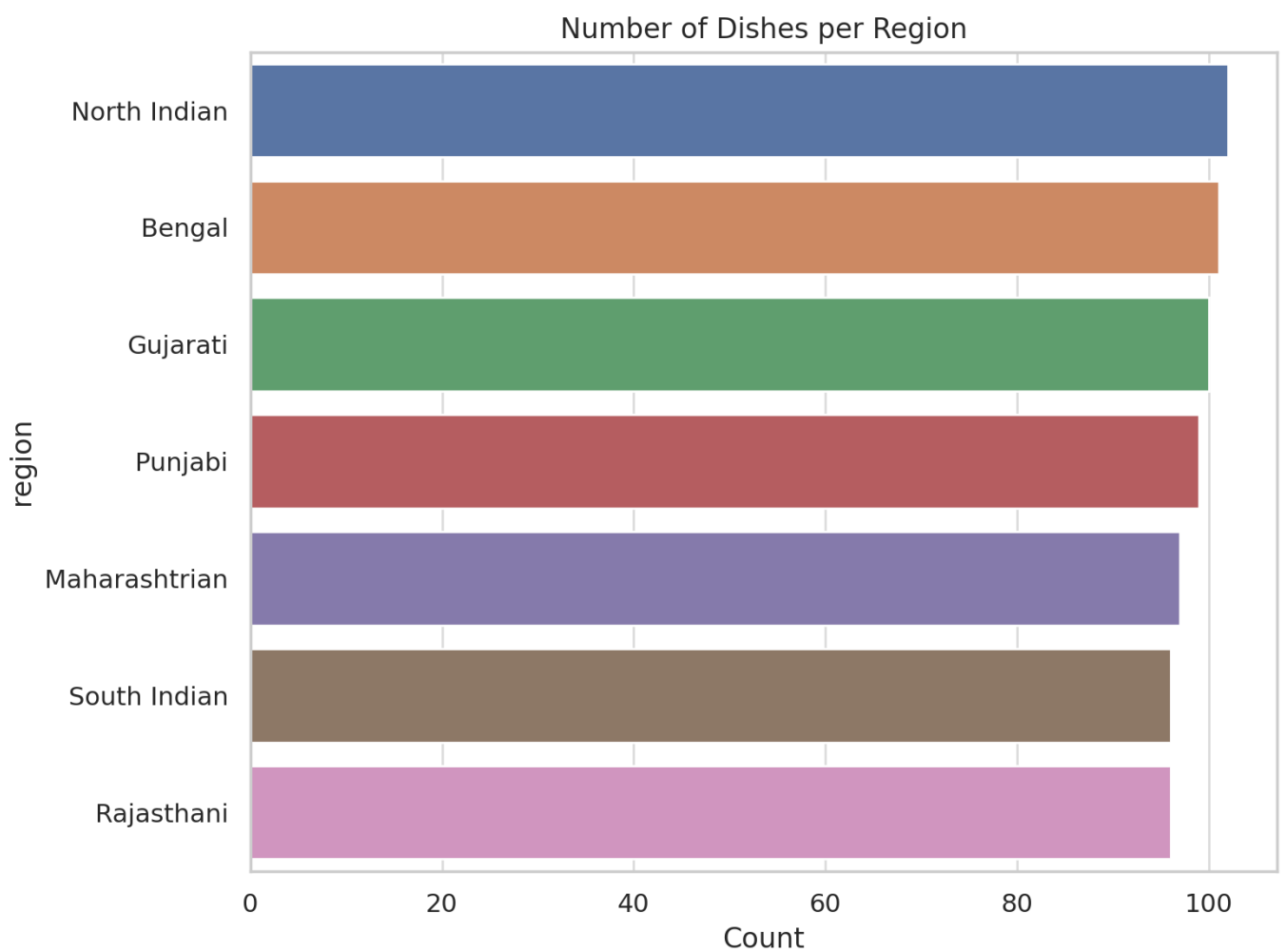


Figure 5.3

Correlation Between Flavors (Heatmap)

Description:

This heatmap displays the Pearson correlation matrix between different flavor dimensions.

Insight:

Positive or negative correlations highlight interdependence. For instance, if sweet and sour are negatively correlated, it may indicate recipes rarely balance both together. High correlation between spicy and salty may reflect common seasoning practices.

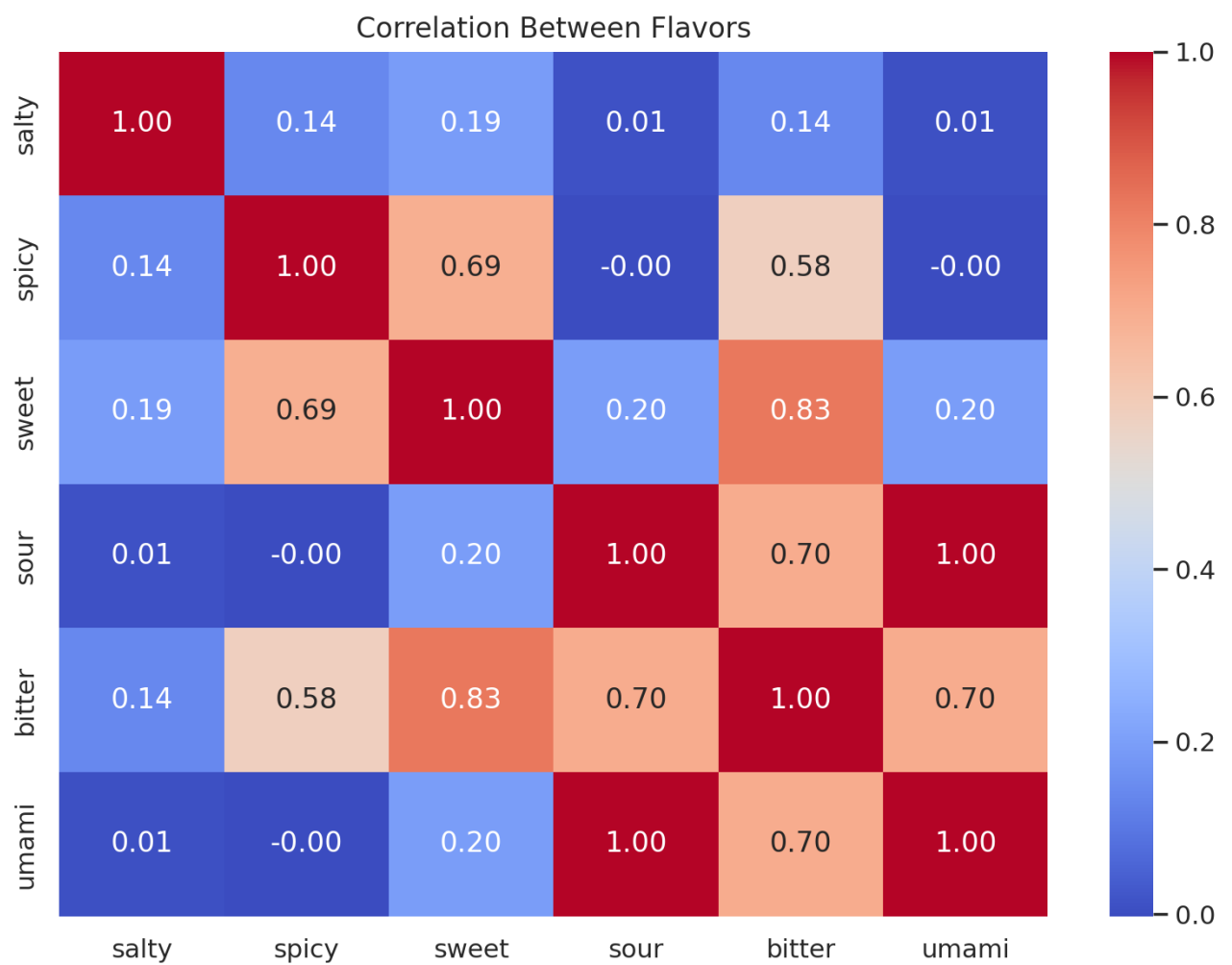


Figure 5.4

5.5 Hypothesis Testing with ANOVA

To test if different Indian regions exhibit distinct ingredient-based flavor trends:

Hypothesis Setup:

- **H₀:** Mean flavor scores are equal across regions.
- **H₁:** At least one region differs in flavor usage.

Method:

One-way ANOVA run on region-wise flavor scores.

Results:

Flavor	F-Statistic	p-value	Interpretation
Spicy	0.79	0.576	Not Significant
Sweet	0.89	0.497	Not Significant
Sour	0.93	0.471	Not Significant
Bitter	0.95	0.458	Not Significant
Salty	0.37	0.891	Not Significant
Umami	0.97	0.439	Not Significant

Table 5.1

Conclusion: No significant flavor usage difference at $p < 0.05$ — suggesting shared culinary bases, possibly due to pan-Indian ingredient availability.

5.6 RAG Pipeline for Recipe Chatbot

A custom Retrieval-Augmented Generation (RAG) pipeline powered the chatbot.

Embedding and Indexing

Mistral 7B model embedded all recipe entries (dish name + ingredients + instructions).

FAISS used to build a high-speed vector index.

Search Logic:

User query (e.g., “suggest a sweet Gujarati dish”) is embedded.

Top-k similar recipes fetched from FAISS.

Retrieved recipes passed as **context** to the language model.

LLM Models Used:

Mistral 7B: Lightweight, fast, and supports retrieval-augmented queries.

LLaMA 2 (tested): Alternative model tested for longer conversations.

RESULTS

This section presents key outcomes derived from both analytical and application-oriented stages of the project.

6.1 Phase 1: Analytical Insights from Recipe Data

- **Ingredient Flavor Mapping:** Successfully mapped over 800 unique ingredients to 6 primary flavor categories (Spicy, Sweet, Sour, Salty, Bitter, Umami) using text mining from *The Flavor Bible* and data scraping from FlavorDB2. This mapping enabled quantitative representation of dish-level flavor profiles.
- **Label Suggestion via Clustering:** Applied Agglomerative Clustering to group semantically similar ingredients and generate suggested_labels for standardized flavor analysis. This step enhanced the consistency and reliability of downstream analysis.
- **ANOVA Analysis Findings:**
Conducted one-way ANOVA to test flavor usage variations across Indian regions.
- **Result:** All six flavors (Spicy, Sweet, Sour, Salty, Bitter, Umami) returned $p\text{-values} > 0.05$, indicating **no statistically significant variation** in flavor usage by region.
- **Interpretation:** While regional identity exists in recipes, the statistical analysis suggests that ingredient-based flavor intensity does not vary significantly across regions — possibly due to lack of quantity/weight measures in the data.

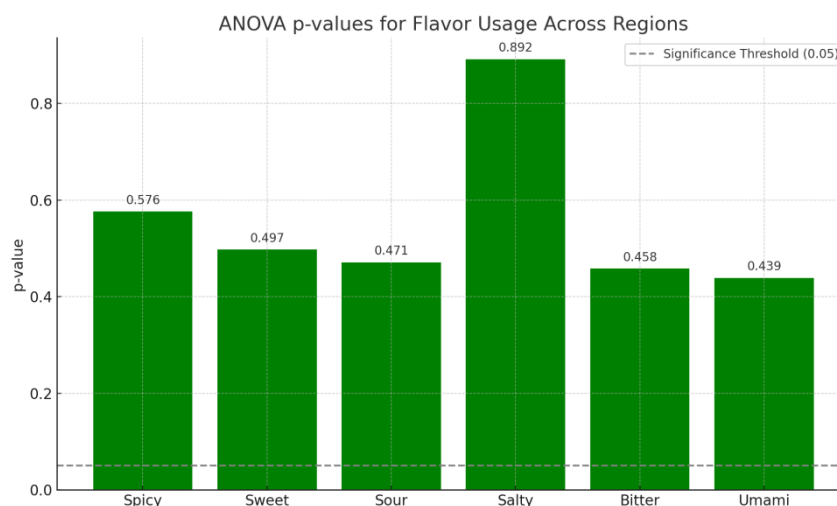


Figure 6.1

6.2 Phase 2: Functional Application via Chatbot

- **Unified Dataset Creation:** Merged data from Tarla Dalal (regional), Hebbbar's Kitchen (generic Indian), and AllRecipes (international) into a homogeneous JSON format with uniform fields: dish_name, region, ingredients, instructions.
- **Chatbot Pipeline:**
 - Used **Mistral 7B** language model integrated through a **RAG (Retrieval-Augmented Generation)** pipeline.
 - Ingredient and recipe data indexed using **FAISS** vector store, enabling semantic similarity-based retrieval.
 - Also tested **LLaMA** models to benchmark conversational accuracy and response variety.
- **Chatbot Capabilities:**
 - Recommends dishes based on available ingredients.
 - Explains regional taste tendencies.
 - Offers recipes with personalized flavor adjustments.
 - Enables comparative regional analysis of flavor profiles.
- **UI Testing:** A minimal interface was tested to allow text input and display chatbot response with recipe images and flavor graph visualizations.

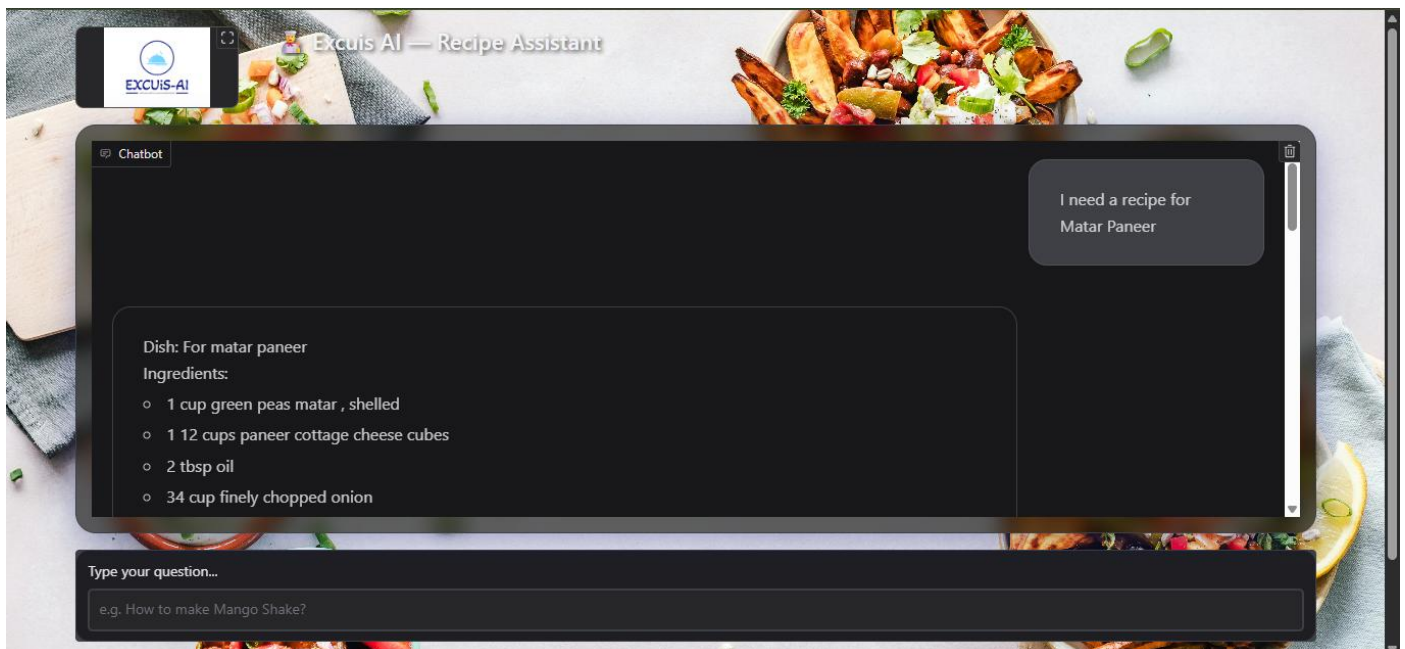


Figure 6.2

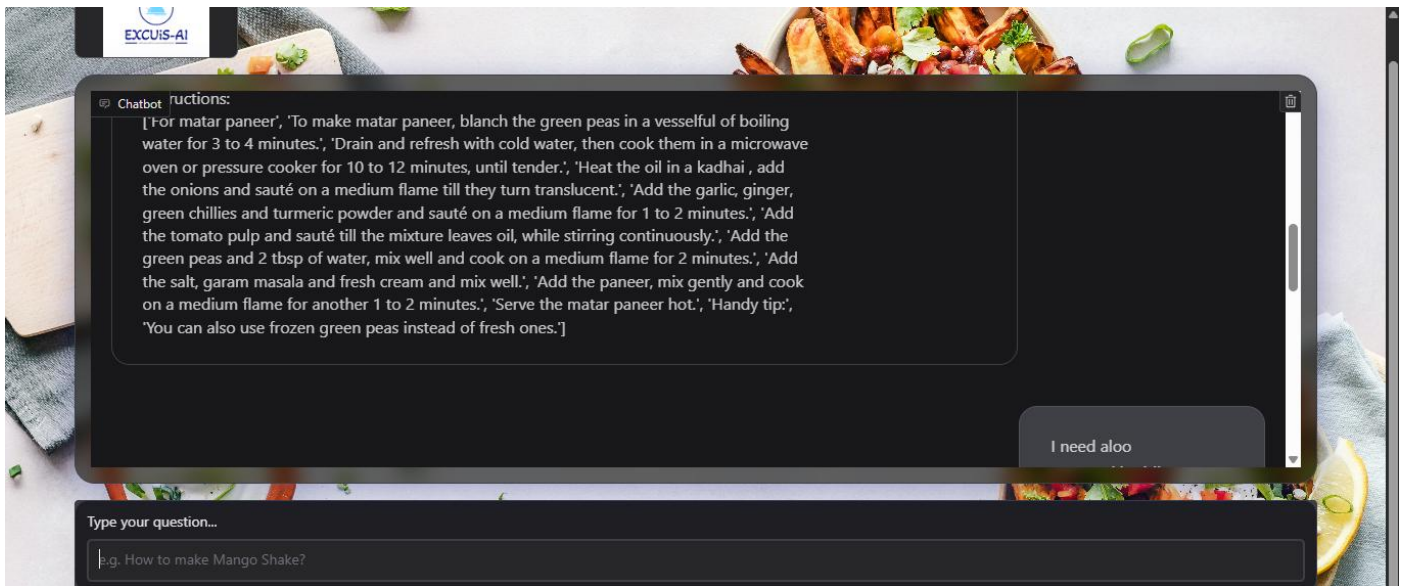


Figure 6.3

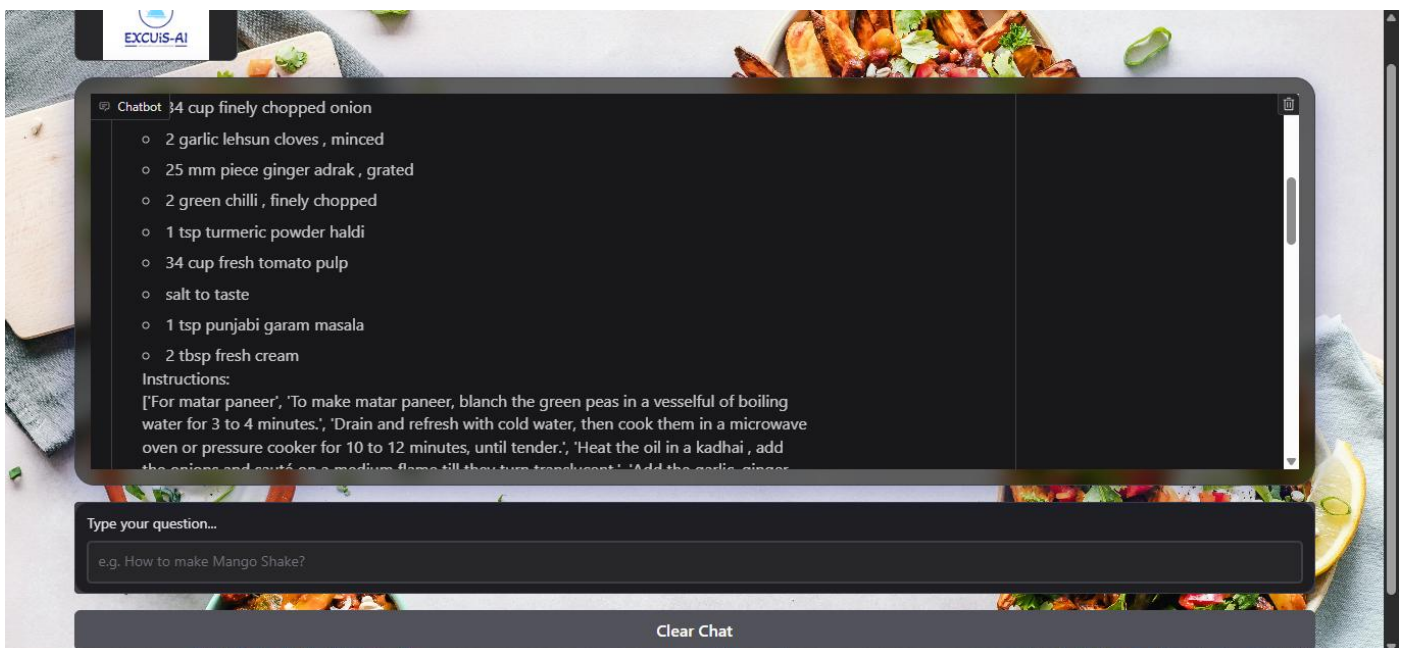


Figure 6.4

CONCLUSION

This project aimed to analyze and understand the regional flavor diversity in Indian cuisine through data-driven methods and build an intelligent, taste-aware recipe chatbot. In Phase 1, we successfully scraped regional recipe data from Tarla Dalal and enriched it with ingredient flavor profiles by integrating data from the Flavor Bible (via text mining) and FlavorDB2 (via scraping). We performed clustering to standardize ingredient labeling and conducted ANOVA hypothesis testing to statistically examine the regional differences in flavor usage. The results revealed no statistically significant difference in the use of salty, spicy, sweet, sour, bitter, or umami ingredients across the Indian regions considered (all p-values > 0.05), indicating a remarkable uniformity in core flavor ingredients despite cultural culinary diversity.

In Phase 2, we broadened the dataset by integrating recipes from Hebbars Kitchen and AllRecipes, followed by structural transformation and region tagging. We then developed a chatbot powered by the Mistral 7B language model, enhanced with a Retrieval-Augmented Generation (RAG) pipeline. Ingredient-to-flavor mappings were stored using FAISS vector indexing, enabling the chatbot to provide flavor-informed responses to user queries. This system allows users to explore regional dishes and receive recommendations aligned with their personal flavor preferences.

Together, the two phases not only showcase an extensive pipeline for culinary data mining and analysis but also demonstrate the power of integrating NLP, clustering, and statistical reasoning to build intelligent food-related applications.

FUTURE ENHANCEMENT

Phase 1 (Flavor Analysis & Modeling)

- **Ingredient Quantity Integration:** Future versions can extract and utilize ingredient quantities to **weight the flavor contributions** more realistically, enabling a true estimation of final dish flavor intensity.
- **Preparation Technique Mining:** Including cooking methods (e.g., roasting, boiling) will refine flavor predictions, as these impact taste significantly.
- **NLP-Based Ingredient Normalization:** More advanced NLP techniques such as transformer-based models can be employed to resolve **synonyms and multi-word entities** with greater precision.
- **Dynamic Region Profiling:** Using unsupervised learning, regions could be clustered **based on actual ingredient usage**, rather than predefined geographic categories.

Phase 2 (Chatbot & Application)

- **Multilingual Support:** Extend the chatbot to support **regional languages** to better serve diverse user demographics in India.
- **Personalized Recipe Recommendations:** Incorporate user feedback and preference learning to generate **custom-tailored meal plans**.
- **Voice Integration:** Enable **voice interaction** with the chatbot using speech-to-text and text-to-speech modules for accessibility.
- **Nutrition and Allergen Awareness:** Augment the chatbot with nutritional data and allergen filtering to support **health-conscious users**.
- **Mobile App Deployment:** Package the solution into a mobile application for real-time, on-the-go usage.

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